

Antony Bush

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Professional Summary

Backend Software Engineer with experience building a multi-stakeholder fintech platform from the ground up — from database schema design and microservice architecture to authentication, API gateway, and regulatory compliance. Proven track record of reducing cloud costs by 23%, boosting service availability by 75%, and delivering real-time communication systems at scale.

Professional Experience

Software Engineer, BondsIndia

Dec 2025 – Present

- **Spearheaded** the ground-up development of a bonds trading platform for users, brokers, and partners — building **14 microservices** in a NestJS/Fastify monorepo with Keycloak SSO (Google, Microsoft) and Kong API Gateway.
- **Engineered** the complete retail KYC journey engine integrating **6 regulatory providers** (DigiLocker, KRA, NSDL/CDSL demat, eSign), handling every edge case to enable **sub-3-minute** user onboarding.
- **Fortified** platform security with **cookie-based auth**, per-frontend **API route isolation** via Kong ensuring microservice-level access control, and PAN-disambiguation sign-in preventing wrong-account access.
- **Optimized** stocksheet rollover and cashflow computation for bond instruments, ensuring **zero data-loss** transitions and accurate yield/price validation across broker integrations.
- **Architected** Security Master APIs across **10+ feature areas** — coupon CRUD with floating-rate triggers, early-exit validation, and issuer ratings — unblocking the entire bonds trading product surface.
- **Designed** the PostgreSQL schema layer with **audit triggers**, FK enforcement, and seed bootstrapping across **55+ migrations** for the greenfield platform.

Software Engineer II, R3 Corda

Sept 2025 – Nov 2025

- **Refactored** Service Account Gateway toward Clean Architecture, decoupling Ktor from the library to make it platform-agnostic and improve testability/maintainability by 30%.
- **Orchestrated** Open Policy Agent authorization, introducing a zero-trust, security-first access model and delivering the first reference integration for Central Bank Digital Currency (CBDC) services, establishing the security standard for platform-wide implementation.

Software Development Engineer I, Jio Platforms Limited

Oct 2023 – Sept 2025

- **Engineered** the migration of business-critical microservices from Python to Node.js, standardizing implementation and achieving a **75% increase in service availability** and an **80% boost in request-handling capacity**.
- **Optimized** media server scalability for JioMeet & JioEvents, enabling **2× participants per node** and **23% cloud cost savings** by tuning SFU scaling logic.
- **Designed** a scalable chat microservice for JioMeet Enterprise, enabling advanced features (threaded replies, sharing) that lifted **user engagement metrics across 1M+ users**.
- **Automated** the complete Media Engine Stack server setup and CI/CD pipeline using Ansible, **slashing deployment time by 60%** and improving release reliability for engineering workflows.
- **Pioneered** MCU capabilities alongside existing SFU architecture, improving stream delivery efficiency by 40% and **reducing client-side bandwidth by up to 60%** for 500+ participant meetings.
- **Overhauled** recording and streaming services with performance optimizations, achieving a **30% reduction in CPU/memory usage** while enhancing stream quality for enterprise clients.
- **Integrated** LiveKit media infrastructure for JioMeet to **support real-time AI media features**, establishing the foundation for advanced AI-driven media processing at scale.

Skills

- **Languages & Frameworks:** TypeScript, JavaScript (Node.js, NestJS, Fastify, Next.js, React)
- **Backend & Systems:** Keycloak, Kong API Gateway, AWS (SES, KMS, SQS), Event-driven Architecture, WebRTC, RabbitMQ
- **DevOps & Cloud:** Docker, Kubernetes, Ansible, Terraform, Jenkins, CI/CD, Nginx, AWS, GCP, Linux, Bash
- **Databases & Caching:** PostgreSQL, MongoDB, Redis, MySQL
- **Monitoring & Observability:** Prometheus, Grafana, Elasticsearch

Education

B.E. in Computer Engineering, Mumbai University

July 2019 – July 2023

- CGPA: 9.67/10.0